Coverage of Children in the American Community Survey Based on California Birth Records

by

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Abstract

The U.S. Census Bureau's American Community Survey (ACS) collects information on individuals and households. The ACS provides survey-based estimates of children drawn from a sample of the U.S. population. However, survey responses may not match administrative records, such as birth records. Birth records should provide a complete account of all births, along with child-parent relationships and demographic characteristics. California is a state that has both a large population of children and a high undercount for young children. This paper uses California as a case study to examine differences between reported versus unreported children in the ACS based on state birth records. Child reporting rates were lower for more recent data years, younger children, for Black and Hispanic mothers, and for more complex households. Child reporting rates were higher for more educated mothers and for households above the poverty line. Using mother's race and Hispanic ethnicity from the birth records combined with poverty indices from the ACS, this analysis also finds that child reporting does not uniformly vary with poverty status across all race and ethnicity groups. This research builds support for the utility of state birth records in analyzing the undercount of children.

Keyword*:* Administrative records, population estimates

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Introduction

The Census Bureau recognizes the well-documented undercount of children in Census Bureau surveys (U.S. Census Bureau 2021a). Undercounting of children can occur when an entire housing unit or household was missed in the survey or census, part of the household was not reported, or only the child was not reported. The undercount of children in the Decennial Census can reduce funding for programs such as Head Start and Children's Health Insurance Programs (CHIP) (Reamer 2019; Murphy and Danielson 2018). To reduce the undercount of children, the Census Bureau has focused research on evaluating the coverage of children in surveys and understanding the causes of undercounting (U.S. Census Bureau 2021a). Unreported children tend to have characteristics that are distinct from those of reported children. Child reporting rates can vary by age, with children under 5 years old having the lowest coverage rates (U.S. Census Bureau 2016). Additionally, coverage rates for children vary by race, with Non-Hispanic White children having the highest coverage rates (O'Hare et al. 2019). More complex household structures are associated with lower coverage rates as well (O'Hare et al. 2019; Fernandez et al. 2018; Jensen et al. 2018).

The Census Bureau's American Community Survey (ACS) collects demographic and economic information on individuals and households, such as age, income, and household structure. The ACS provides survey-based estimates of children drawn from a sample of the U.S. population. However, survey responses may not match administrative records, such as birth records. Birth records are assumed to provide a virtually complete account of all births, along with child-parent relationships and demographic characteristics. California has both the largest share of the population of U.S. children and a known high undercount rate for young children (U.S. Census

Bureau 2022; O'Hare et al. 2019; O'Hare 2014b). This paper uses California as a case study to examine differences between reported versus unreported children in a Census Bureau survey based on state birth records.

Data

This analysis used California birth records, the ACS, the Numerical Identification file (Numident), California infant mortality records, and the Census Household Composition Key. The Census Bureau's Person Identification Validation System (PVS) was used to assign individual Protected Identification Keys (PIKs) using a probabilistic matching algorithm and personally identifiable information (PII) (Wagner and Layne 2014). This paper analyzes the linkages between the American Community Survey and the California birth records using PIKs.

California birth records

The California birth records used in this analysis span from 1989 to 2019. 98% of the children and 84% of the mothers in this sample received a PIK (Table 1). The PIK rate for fathers is much lower at 41% due to insufficient personally identifiable information for fathers in the birth records before 2005. The birth records from 1989 to 1996 were not assigned any father PIKs. For the birth records from 1997 to 2004, the father's first name was not provided on the birth records and PIK rates were lower than later years beginning in 2005 when this information was provided.

Table 1. PIK Rate of All California Birth Records by Household Role (1989-2019)

Household Role	Assigned PIK	Not Assigned PIK	Total
Child	16,220,000 98%	324,000 2%	16,540,000
Mother	13,950,000 84%	2,599,000 16%	16,550,000
Father	6,716,000 41%	9,832,000 59%	16,550,000

The following data from the birth records were used to characterize reported and unreported children, based on factors known to be associated with the undercounting of children[1]:

- Child's birth date, to calculate child ages[2] (U.S. Census Bureau 2021a; Griffin and O'Hare 2020)
- Mother's birth country (Johnson 2022; Fernandez et al. 2018)
- Mother's race and Hispanic ethnicity (O'Hare et al. 2019; O'Hare et al. 2016)
- Mother's high school completion status as recorded on the birth record (Fernandez et al. 2018; O'Hare 2014b)

Death data

The Numident file is an administrative record of interactions with the Social Security Administration and contains reported death date for individuals (Finlay and Genadek 2021). The Numident file was linked to the California birth records by PIK. Additionally, infant mortality

[1] While this analysis characterizes children based on the mother's data from the birth records, the results still include children who were not reported as living with their mother in the ACS so as not to exclude children whose mothers did not receive PIKs.

[2] Age misreporting could affect this analysis because it would change the number of children reported in each age group (Fernandez et al. 2018). Given the known significant undercount of children under 5 years old and to avoid age misreporting from ACS responses, I calculated child age at the time of ACS response using the birthdate from their birth record.

data from the California Department of Health Care Access and Information's Linked Birth File for children born in 1991 to 2011 were used. Birth records were excluded from this analysis when the recorded death date for the child in the Numident or infant mortality records occurred before the expected ACS response date, as those children should not be reported on the ACS.

Census Household Composition Key

The Census Household Composition Key (CHCK) is a Census Bureau dataset that links child and parent PIKs using the Numident data received from the Social Security Administration, beginning with births in 1997 (Genadek et al. 2021; Luque and Wagner 2015). Each CHCK vintage includes individuals aged 0-19. The California birth records were linked to the Census Household Composition Key data files by child PIK. When a parent's PIK was missing from the birth records but appeared in the CHCK, the parent PIK from the CHCK was linked to the child record.[3]

American Community Survey

American Community Survey data from 2005 to 2019 was used in this analysis. All ACS records with assigned PIKs were used for linkage to the California birth records, regardless of state, to include children who were born in California but lived in another state at the time of ACS response. Over 99% of the ACS records were assigned PIKs. The following data from the ACS

[3] Genadek et al. (2021) found that an average of 94.5% of all births were linked to at least one parent in the CHCK for annual birth cohorts from 1997 to 2016 in the 2016-2019 CHCK files.

were used to characterize reported and unreported children, based on factors known to be associated with the undercounting of children:

- Ease of speaking English (Walejko et al. 2019; Fernandez et al. 2018)
- Presence of subfamilies and non-related persons in the household, as indicators of household complexity (Griffin and O'Hare 2020; Jensen et al. 2018)
- Poverty status (Fernandez et al. 2018)
- Response mode (O'Hare et al. 2019)

Limitations

PIKs may not be assigned when there is insufficient PII in the dataset or when the PII available is not a unique match to the administrative records used in the PVS process (Bond et al. 2014). Because the sample in this analysis is limited to records that can be linked by PIK, the results are subject to bias from non-random inability to assign PIKs to records. Young children, racial and ethnic minorities, immigrants, and low-income individuals are all less likely to receive a PIK and more likely to be undercounted in surveys (Johnson 2022; O'Hare et al. 2019; Fernandez et al. 2018; Bond et al. 2014).

This analysis may underestimate the number of unreported children in cases where the child did not receive a PIK in the birth records and was unreported in the ACS, because these children were excluded from the sample entirely. Additionally, some children in this analysis who received a PIK in the birth records may be mislabeled as unreported if they were reported in their ACS household but were not assigned a PIK in the ACS data.

Finally, this analysis relies on the assumption that a child should be reported in their parents' household. Children are labeled as unreported in this analysis when at least one parent was linked to an ACS response, but the child was not linked to any ACS response. However, this characterization of a child as unreported is inaccurate if the child does not live with at least one parent and the household where the child lives was not included in the ACS sample. About 4.6% of children in California lived with neither parent in 2018 (Population Reference Bureau 2019). Children who live with neither parent can vary from other children by demographic characteristics. Nationally, children who live with neither parent are more likely to be older and Black Non-Hispanic or Hispanic of any race (Hemez and Washington 2021).

Methods

Children, mothers, and fathers from the California birth records were linked to ACS records by PIK. The birth records and ACS responses included in this analysis are only those where at least one birth record PIK (child, mother, and/or father PIK) was linked to an ACS record. Less than 1% of the linked records had duplicate responses in the ACS. In these cases, I kept the record with the earliest response date.

This linkage created two groups of children:

- Children considered reported, where the child PIK from the birth records was matched to an ACS record. Children in this group are considered reported regardless of whether a parent was in the same household, so as not to exclude children who were reported but do not live with a parent.

- Children considered not reported, where at least one parent PIK from the birth records was matched to an ACS record, but the child's PIK was not matched to an ACS record. Children in this group are assumed to be living with at least one parent in their household but not reported.

After this linkage, the sample in this analysis totaled 1.89 million records. As shown in Table 2, 96% of the children in the sample had assigned PIKs for at least one parent from the birth records and CHCK. These children may fall into the reported group if they were matched to an ACS record by PIK, or into the unreported group if their parent was matched to an ACS record by PIK but the child was not matched. Four percent of children did not have assigned PIKs for either parent. Based on the methods of this analysis described above, these children were reported in the ACS and linked to the birth records by PIK because it was not possible to link their parents to any ACS record.

Table 2. Parent Linkage for All Children in Sample

Parent Linkage	Count	Percent of All Children in Sample
Both parents assigned PIKs	1,224,000	65%
Only mother assigned PIK	525,000	28%
Only father assigned PIK	59,000	3%
Neither parent assigned PIK	83,000	4%
Total	1,891,000	100%

As noted above, demographic and economic ACS data were used in this analysis to characterize reported and unreported children. If a child was linked to an ACS record, the responses from the child's ACS record were used. If the mother was linked to an ACS record but the child was not reported, the responses from the mother's ACS record were used, based on the assumption that children are more likely to live with their mother (Anderson et al. 2022). Finally, if the father

was linked to an ACS record but the mother and child were not found, the responses from the father's ACS record were used.

Analysis

The following sections describe PIK match rates and characterize child reporting by age, mother's demographic characteristics, English language proficiency, household complexity, collection mode, and household poverty status. Cells in the tables below may be non-additive due to rounding.

Match Rates

Table 3 shows child PIK match rates for each ACS year. Between 12% and 21% of children from the birth records were not reported when at least one parent had an ACS record each year. The share of children reported in the ACS in this analysis decreased by 8 percentage points from 2005 to 2019. This decrease in child reporting over time follows the growing undercount of young children since 1980 (O'Hare 2014a).

Table 3. California Birth Records and ACS Match Rates

ACS Year	Children Reported	Children Not Reported
2005	103,000 88%	14,000 12%
2006	105,000 86%	17,500 14%
2007	101,000 85%	18,000 15%
2008	98,000 84%	19,000 16%
2009	94,500 81%	21,500 19%
2010	100,000 84%	18,500 16%
2011	105,000 83%	21,500 17%
2012	120,000 82%	25,500 18%
2013	111,000 82%	25,000 18%
2014	115,000 81%	27,500 19%
2015	112,000 81%	27,000 19%
2016	105,000 80%	27,000 20%
2017	99,000 79%	26,500 21%
2018	97,000 80%	24,500 20%
2019	90,500 80%	22,000 20%

Age

Children were categorized into the following age groups: 0 to 4, 5 to 9, 10 to 14, and 15 to 17. Each child's age was calculated based on their birthdate from the birth records and the ACS response date. The birthdate from the birth records was used in this calculation to avoid using

misreported birthdates in the ACS for reported children and as the only birthdate source for unreported children. For children reported in the ACS, 85% of ACS records had the same birthdate as the birth record and 90% had the same reported age as calculated using the birth record birthdate and the child's ACS response date. The parent response date was used for calculating the ages of unreported children.

Table 4 describes each age group's share of the total and child reporting rates by age group. Each age group makes up between 18% to 29% of the total sample. The 15-17 age group was expected to be a slightly smaller share of the total sample because the age group only includes 3 different ages versus the 4 different ages in each other age group. A greater share of children in each age group were reported in the ACS at older ages, increasing from 80% of children under 5 years to 86% of children between ages 15 and 17. These results align with the known higher undercount for young children (age 0-4) than for older children (O'Hare et al. 2019; U.S. Census Bureau 2016).

Table 4. Child Reporting Rates by Age Group

Age Group	Share of Total Sample	Child Reported	Child Not Reported
Age 0-4	476,000 25%	380,000 80%	96,000 20%
Age 5-9	526,000 28%	426,000 81%	100,000 19%
Age 10-14	548,000 29%	458,000 84%	90,000 16%
Age 15-17	340,000 18%	291,000 86%	48,500 14%

Mother's Birthplace

Table 5 describes the sample and child reporting rates by whether a mother was born in the United States according to the birth records. 59% of mothers in the sample were born in the

United States. Less than 1% of mothers were missing this data. 84% of children of US-born mothers were reported in the ACS, while children of foreign-born mothers were reported at a slightly lower rate of 80%. While these results match prior research where having a foreign-born parent is associated with the undercounting of children such as O'Hare et al. (2020), this analysis does not account for emigration by native-born children of foreign-born parents.

Table 5. Child Reporting Rates by Mother's Birthplace

Mother's Birthplace	Share of Total Sample	Children Reported	Children Not Reported
US-Born	1,116,000 59%	936,000 84%	180,000 16%
Foreign-Born	773,000 41%	618,000 80%	155,000 20%
Missing Data	1,600 0%	1,400 85%	250 15%

Mother's Race and Hispanic Ethnicity

The mother's race and ethnicity information from the birth records was categorized into the following combinations: Non-Hispanic White, Non-Hispanic Asian, Non-Hispanic Black, Non-Hispanic Other Race, and Hispanic of any race. When race was reported but not Hispanic ethnicity, the mother was assumed to be Non-Hispanic. Notably, 46% of the mothers in the sample are Hispanic of any race (Table 6). Prior studies have found that undercounting is higher for Hispanic and Non-Hispanic Black children (O'Hare et al. 2019; O'Hare et al. 2016). Child reporting is lowest among children of Hispanic mothers and Non-Hispanic Black mothers in this analysis as well, with 76% of these children reported in each group compared to 89% of Non-Hispanic White children.

Table 6. Child Reporting Rates by Mother's Race/Ethnicity

Race/Ethnicity	Share of Total Sample	Child Reported	Child Not Reported
Non-Hispanic White	658,000 35%	584,000 89%	74,000 11%
Non-Hispanic Asian	220,000 12%	197,000 90%	23,000 10%
Non-Hispanic Black	92,500 5%	70,500 76%	22,000 24%
Non-Hispanic Other Race	40,500 2%	32,500 80%	8,100 20%
Hispanic, Any Race	861,000 46%	656,000 76%	205,000 24%
Missing	19,500 1%	16,500 84%	3,100 16%

Mother's Education

The California birth records include whether a mother completed high school. Table 7 shows each education group's share of the total sample, with 72% of the mothers in the sample having completed high school. Children of mothers who completed high school were reported in 86% of cases, while children of mothers who did not complete high school were only reported in 73% of cases, matching prior research where lower educations level were associated with the undercounting of children (Fernandez et al. 2018; O'Hare 2014b).

Table 7. Child Reporting Rates by Mother's Education

Mother's Education	Share of Total Sample	Child Reported	Child Not Reported
Less Than High School	477,000 25%	347,000 73%	130,000 27%
High School Completion	1,363,000 72%	1,166,000 86%	197,000 14%
Missing Data	51,500 3%	43,500 84%	8,200 16%

Ease of Speaking English

Table 8 uses data from the ACS on the ease of speaking English for individuals 5 years old or older. When a child under 5 years old was reported in the ACS, ease of speaking English data were taken from a parent's ACS response where possible. Eighty-five percent of the sample in this analysis responded that they spoke English very well. From the group with the greatest English-speaking ability ("very well") to the group with the lowest English-speaking ability ("not at all"), child reporting rates decreased by 48 percentage points. These results are similar to those of Fernandez et al. (2018), which found that children in households with low English language proficiency were less likely to be matched to the Decennial Census.

Table 8. Child Reporting Rates by Ease of Speaking English

Ease of Speaking English	Share of Total Sample	Child Reported	Child Not Reported
Very Well	1,644,000 85%	1,373,000 84%	271,000 16%
Well	141,000 7%	101,000 72%	40,000 28%
Not Well	78,000 4%	41,000 53%	37,000 47%
Not At All	29,000 2%	10,500 36%	18,500 64%
Missing Data	31,500 2%	31,000 99%	400 1%

Table 9 describes child reporting rates by combinations of mother's race and Hispanic ethnicity from the birth records and English-speaking ability from the ACS. Across races, child reporting rates tend to decrease as English-speaking ability decreases. However, child reporting rates tend to drop more dramatically for children of mothers who are Hispanic of any race, decreasing 45 percentage points from those who speak English very well to those who do not speak English at all. Child reporting rates drop similarly for children of Non-Hispanic Black mothers, but the

number of records for these mothers who do not speak English at all is small (N=70). In contrast, child reporting only decreases by 29 percentage points for children of Non-Hispanic Asian mothers and by 38 percentage points for children of Non-Hispanic White mothers.

Table 9. Child Reporting Rates by Ease of Speaking English and Mother's Race/Ethnicity

	Non-Hispanic White		Non-Hispanic Asian		Non-Hispanic Black		Non-Hispanic Other Race		Hispanic, Any Race	
	Child Reported	Child Not Reported	Child Reported	Child Not Reported	Child Reported	Child Not Reported	Child Reported	Child Not Reported	Child Reported	Child Not Reported
Very Well	576,000 88%	81,000 12%	165,000 91%	17,000 9%	68,500 75%	23,000 25%	31,000 78%	8,500 22%	522,000 79%	140,000 21%
Well	6,000 77%	1,800 23%	22,000 81%	5,100 19%	650 68%	300 32%	700 67%	350 33%	70,500 69%	32,000 31%
Not Well	2,800 74%	1,000 26%	8,400 75%	2,800 25%	300 67%	150 33%	300 67%	150 33%	29,000 47%	33,000 53%
Not At All	300 50%	300 50%	800 62%	500 38%	20 29%	50 71%	60 55%	50 45%	9,200 34%	17,500 66%

Presence of Subfamilies

A subfamily is defined as "a married couple with or without never-married children under 18 years old, or one parent with one or more never-married children under 18 years old ... in a household where the householder or householder's spouse is a relative" (U.S. Census Bureau 2021b). Survey respondents may be less likely to count children in households with subfamilies (Griffin and O'Hare 2020). Table 10 describes group shares and child reporting by whether subfamilies were reported in the ACS. In 86% of the records in this analysis, subfamilies were not reported in the ACS household. Child reporting among households without subfamilies and households with subfamilies were similar, at 84% and 82% respectively. Children in households who were missing this data were only reported in 11% of cases.

Table 10. Child Reporting by Presence of Subfamilies

Presence of Subfamilies	Share of Total Sample	Child Reported	Child Not Reported
No Subfamilies in Household	1,629,000 86%	1,366,000 84%	263,000 16%
Subfamilies in Household	226,000 12%	186,000 82%	40,500 18%
Missing Data	35,000 2%	3,700 11%	31,500 89%

Presence of Non-Related Persons

A non-related person is defined as "any household member, including foster children, not related to the householder by birth, marriage, or adoption" (U.S. Census Bureau 2021b). This category includes individuals such as roommates and unmarried partners. Eighty-five percent of the records in the sample did not report non-related persons in the household in the ACS (Table 11). In contrast to child reporting by presence of subfamilies, households without non-related persons reported children 85% of the time, while households with non-related persons in the household

only reported children 75% of the time. These results may indicate differences in child reporting specifically when non-related persons are in the household, aligning with results from Jensen et al. (2018). As in Table 10 for the presence of subfamilies, children in households who were missing non-related persons data were only reported in 11% of cases.

Table 11. Child Reporting by Presence of Non-Related Persons

Presence of Non-Related Persons	Share of Total Sample	Child Reported	Child Not Reported
No Non-Related Persons in Household	1,602,000 85%	1,363,000 85%	239,000 15%
Non-Related Persons in Household	253,000 13%	189,000 75%	64,000 25%
Missing Data	35,000 2%	3,700 11%	31,500 89%

Collection Mode

Table 12 describes the shares of the sample by ACS collection mode. The greatest shares of responses were by mail (36%), computer assisted personal interview (CAPI) (29%), and internet (23%). Group quarters personal visits made up the smallest share of the total at 2% of all responses in the sample. Child reporting rates were highest for mail responses (88%) and internet responses (87%), matching prior findings that children in households enumerated by self-response are more likely to be counted (O'Hare et al. 2019). Child reporting rates were low for group quarters personal visit responses (11%). This is likely due to parents living in group quarters where children are not expected to live with parents. Nonetheless, these group quarters were included in the sample because some children live in group quarters intended to house adults, such as jails (Zeng 2018).

Table 12. Child Reporting by Collection Mode

Collection Mode	Share of Total Sample	Child Reported	Child Not Reported
Mail	690,000 36%	604,000 88%	85,500 12%
Computer Assisted Telephone Interview (CATI)	185,000 10%	154,000 83%	31,000 17%
Computer Assisted Personal Interview (CAPI)	555,000 29%	423,000 76%	132,000 24%
Group Quarters Personal Visit	35,500 2%	3,800 11%	31,500 89%
Internet	427,000 23%	372,000 87%	55,000 13%

Poverty

Table 13 describes the sample and child reporting rates by whether the household was in poverty in the ACS. Eighty-one percent of the sample was not in poverty and 17% of the sample was in poverty. Two percent of the sample was missing data on poverty status, which was primarily from group quarters responses. Child reporting was 5 percentage points lower for households in poverty than for families not in poverty.

Table 13. Child Reporting by Poverty Status

Poverty	Share of Total Sample	Child Reported	Child Not Reported
Not in Poverty	1,530,000 81%	1,289,000 84%	241,000 16%
In Poverty	324,000 17%	257,000 79%	67,000 21%
Missing Data	36,500 2%	9,600 26%	27,000 74%

Table 14 further stratifies households into the following categories: 0-100% of the poverty line, 101-200% of the poverty line, 201-300% of the poverty line, 301-400% of the poverty line, and 401% of the poverty line and above. The greatest share of households come from the group at or

above 401% of the poverty line (31%). Child reporting rose from 79% for families in poverty to 87% for households in the group at or above 401% of the poverty line. These results align with those of Fernandez et al. (2018), which found that children living in households in the groups below 200% of the federal poverty line were less likely to be found in the Decennial Census.

Table 14. Child Reporting by Household Percentage of the Poverty Line

Poverty Status	Share of Total Sample	Child Reported	Child Not Reported
Up to 100% of Poverty Line	329,000 17%	261,000 79%	68,000 21%
101-200% of Poverty Line	404,000 21%	331,000 82%	73,000 18%
201-300% of Poverty Line	308,000 16%	252,000 82%	55,500 18%
301-400% of Poverty Line	222,000 12%	185,000 83%	37,000 17%
401% of Poverty Line and Above	592,000 31%	518,000 87%	74,500 13%
Missing Data	36,500 2%	9,600 26%	27,000 74%

Table 15 additionally characterizes child reporting by poverty status from the ACS and mother's race and Hispanic ethnicity from the birth records. Child reporting increases for children of Non-Hispanic White mothers by 13 percentage points and for children of Non-Hispanic Asian mothers by 7 percentage points from the group in poverty to the group at or above 401% of the poverty line. However, child reporting does not increase notably for children of Non-Hispanic Black mothers and decreases by 6 percentage points for children of Hispanic mothers. Therefore, the relationship between poverty and child reporting may not be uniform across race groups.

Table 15. Child Reporting by Household Percentage of the Poverty Line and Mother's Race

	Non-Hispanic White		Non-Hispanic Asian		Non-Hispanic Black		Non-Hispanic Other Race		Hispanic, Any Race	
	Child Reported	Child Not Reported	Child Reported	Child Not Reported	Child Reported	Child Not Reported	Child Reported	Child Not Reported	Child Reported	Child Not Reported
Up to 100% of Poverty Line	51,000 79%	13,500 21%	17,500 85%	2,900 15%	20,500 78%	5,700 22%	5,800 76%	1,800 24%	164,000 79%	43,500 21%
101-200% of Poverty Line	74,500 87%	11,000 13%	26,500 88%	3,600 12%	16,000 81%	3,600 19%	6,500 82%	1,400 18%	205,000 80%	52,500 20%
201-300% of Poverty Line	83,000 89%	10,000 11%	26,500 89%	3,400 11%	11,000 80%	2,600 20%	5,300 82%	1,200 18%	124,000 77%	38,000 23%
301-400% of Poverty Line	80,000 91%	8,200 9%	25,000 89%	2,900 11%	7,600 81%	1,800 19%	4,000 84%	800 16%	67,000 74%	23,000 26%
401% of Poverty Line and Greater	293,000 92%	25,000 8%	101,000 92%	9,200 8%	14,500 79%	3,700 21%	10,500 85%	1,800 15%	91,000 73%	33,500 27%

Conclusion

This paper takes advantage of administrative records from California, which has both a large population of children and a high undercount for young children, to characterize undercounted children. Patterns of child reporting in this analysis generally follow those of prior research, offering further support for the characteristics known to be associated with the undercounting of children, such as younger child age, race and Hispanic ethnicity, English proficiency, and household complexity. Moreover, this analysis provides support for using state birth records to supplement demographic analyses of undercounting.

Given the noticeable variations in child reporting by poverty status and mother's race and ethnicity, future research should take advantage of other administrative records housed at the Census Bureau, such as 1040 and W2 data from the Internal Revenue Service (IRS). Use of IRS income data could further refine the analysis by using IRS reported income rather than survey-reported income and by examining child reporting rates for children who were not claimed as dependents by their parents. Additionally, this analysis can be expanded by linking birth records to the Decennial Census, which is intended to have a complete count of all children in the United States.

References

Anderson, L. R., Hemez, P.F., and Kreider, R.M. (2022). "Living Arrangements of Children: 2019." *Current Population Reports* No. P70-174. U.S. Census Bureau. https://www.census.gov/content/dam/Census/library/publications/2022/demo/p70-174.pdf

Bond, B.J., Brown, D., Luque, A., and O'Hara, A. (2014). "The Nature of the Bias When Studying Only Linkable Person Records: Evidence from the American Community Survey." *U.S. Census Bureau Working Paper* No. CARRA-WP-2014-08. https://www.census.gov/content/dam/Census/library/working-papers/2014/adrm/carra-wp-2014-08.pdf

Fernandez, L., Shattuck, R., and Noon, J. (2018). "The Use of Administrative Records and the American Community Survey to Study the Characteristics of Undercounted Young Children in the 2010 Census." *U.S. Census Bureau Working Paper* No. CARRA-WP-2018-05. https://www.census.gov/content/dam/Census/library/working-papers/ 2018/adrm/carra-wp-2018-05.pdf

Finlay, K., and Genadek, K.R. (2021). "Measuring All-Cause Mortality with the Census Numident File." *U.S. Census Bureau Working Paper* No. ADEP-WP-2021-03. https://www.census.gov/content/dam/Census/library/working-papers/2021/econ/wp-measuring-all-cause-mortality.pdf

Genadek, K.R., Sanders, J., and Stevenson, A.J. (2021). "Measuring U.S. Fertility using Administrative Data from the Census Bureau." *U.S. Census Bureau Working Paper* No. ADEP-WP2021-02. https://www.census.gov/content/dam/Census/library/working-papers/2021/econ/chck-adep-working-paper.pdf

Griffin, D.H., and O'Hare, W.P. (2020). “Are Census Omissions of Young Children Due to Respondent Misconceptions about the Census?” *Int'l J. Soc. Sci. Stud., 8, 59.* https://doi.org/10.11114/ijsss.v8i6.4994

Hemez, P., & Washington, C. (2021, April 12). *Number of Children Living Only With Their Mothers Has Doubled in Past 50 Years*. U.S. Census Bureau. https://www.census.gov/library/stories/2021/04/number-of-children-living-only-with-their-mothers-has-doubled-in-past-50-years.html

Jensen, E., Schwede, L., Griffin, D., and Konicki, S. (2018). “Investigating the 2010 Undercount of Young Children – Analysis of Complex Households.” *U.S. Census Bureau Center for Behavioral Science Methods Research Report Series, Survey Methodology* No. 2018-20. http://www.census.gov/content/dam/Census/library/working-papers/2018/adrm/rsm201820.pdf

Johnson, J. E. (2022). “Does the census miss the native-born children of immigrant mothers? Evidence from state-level undercount by race and Hispanic status.” *Population Research and Policy Review*, *41*(1), 139-195.

Luque, A., and Wagner, D. (2015). “Assessing coverage and quality of the 2007 Prototype Census Kidlink Database.” *U.S. Census Bureau Working Paper* No. CARRA-WP-2015-07. https://www.census.gov/content/dam/Census/library/working-papers/2015/adrm/carra-wp2015-07.pdf

Mulrow, E., Ali, M., Pramanik, S., and Fontes, A. (2011). “Assessment of the U.S. Census Bureau’s Person Identification Validation System.” NORC. https://www.norc.org/PDFs/May%202011%20Personal%20Validation%20and%20Entity

%20Resolution%20Conference/PVS%20Assessment%20Report%20FINAL%20JULY%202011.pdf

Murphy, P., & Danielson, C. (2018). "Census-related funding in California. Public Policy Institute of California." https://www.gcir.org/sites/default/files/resources/PPIC-census-related-funding.pdf

O'Hare, W. P. (2014a). "Historical Examination of Net Coverage Error for Children in the U.S. Decennial Census: 1950 to 2010." *U.S. Census Bureau Working Paper* No. SSM2014-03. https://www.census.gov/content/dam/Census/library/working-papers/2014/adrm/ssm2014-03.pdf

O'Hare, W. P. (2014b). "State-level 2010 census coverage rates for young children." *Population Research and Policy Review*, 33, 797–816.

O'Hare, W. P., Mayol-Garcia, Y., Wildsmith, E., & Torres, A. (2016). "The invisible ones: How Latino children are left out of our nation's census count." *Report, Child Trends Hispanic Institute and the National Association of Latino Elected Officials*. https://cms.childtrends.org/wp-content/uploads/2016/04/2016-16TheInvisibleOnesLatinoCensus.pdf

O'Hare, W. P., Griffin, D., and Konicki, S. (2019). "Investigating the 2010 Undercount of Young Children – Summary of Recent Research." https://www2.census.gov/programs-surveys/decennial/2020/program-management/final-analysis-reports/2020-report-2010-undercount-children-summary-recent-research.pdf

O'Hare, W. P., Jacobsen, L.A., Mather, M., and VanOrman A. (2020). "Predicting Tract-Level Net Undercount Risk for Young Children." Population Reference Bureau. https://www.prb.org/wp-content/uploads/2020/12/us-census-undercount-of-children-

1.pdf

Population Reference Bureau. (2019). “Living Arrangement for Children, by Presence of Parents.” https://www.kidsdata.org/topic/41/living-with-parents/table

Reamer, A. (2019). “Fifty-five Large Federal Census-guided Spending Programs: Distribution by State. Counting for Dollars 2020: Report #5.” The George Washington University. https://gwipp.gwu.edu/sites/g/files/zaxdzs611/files/downloads/Counting%20Dollars%20Brief%20%235%20May%202019.pdf

U.S. Census Bureau. (2016). “Investigating the 2010 Undercount of Young Children – A New Look at 2010 Census Omissions by Age.” https://www2.census.gov/programs-surveys/decennial/2020/program-management/memo-series/2020-report-2010-undercount-children-ommissions.pdf

U.S. Census Bureau. (2021a). “The Undercount of Young Children.” https://www.census.gov/programs-surveys/decennial-census/decade/2020/planning-management/plan/undercount-of-young-children.html

U.S. Census Bureau. (2021b). “American Community Survey and Puerto Rico Community Survey.” https://www2.census.gov/programs-surveys/acs/tech_docs/subject_definitions/2021_ACSSubjectDefinitions.pdf

U.S. Census Bureau. (2022). *2021 American Community Survey 1-Year Estimates Subject Tables*. https://data.census.gov/table?q=children&g=010XX00US$0400000&tid=ACSST1Y2021.S0901&tp=true

U.S. Census Bureau. (2023). *People and Households Represented in Each American Community Survey Data Collection Mode.* https://www.census.gov/library/visualizations/interactive/acs-collection.html

Wagner, D., and Layne, M. (2014). “The person identification validation system (PVS): applying the Center for Administrative Records Research and Applications’(CARRA) record linkage software.” *U.S. Census Bureau Working Paper* No. CARRA-WP-2014-01 https://www.census.gov/content/dam/Census/library/working-papers/2014/adrm/carrawp-2014-01.pdf

Walejko, G., Shia, L., Scheid, S., & Griffin, D. (2019). “Researching the Attitudes of Households Reporting Young Children—a Summary of Results from the 2020 Census Barriers, Attitudes, and Motivators Study (CBAMS) Survey.” US Census Bureau. https://www2.census.gov/programs-surveys/decennial/2020/program-management/final-analysis-reports/2020-report-cbams-attitudes-reporting-children.pdf

Zeng, Z. (2018). “Jail Inmates in 2016.” Bureau of Justice Statistics Bulletin NCJ 251210. https://bjs.ojp.gov/content/pub/pdf/ji16.pdf